



YISHUN INNOVA JUNIOR COLLEGE  
2026 JC1 MOCK PROMOTIONAL EXAMINATION  
**Higher 2**

CANDIDATE  
NAME

CG

INDEX NO

---

**COMPUTING**

Paper 1 Written

**9569/01**

**2026**

**1 hour 10 minutes**

Additional Materials: 12-page writing booklet

---

**READ THESE INSTRUCTIONS FIRST**

An answer booklet will be provided with the question paper.

You should follow the instructions on the front cover of the answer booklet.

If you need additional answer paper ask the invigilator for a continuation booklet.

Answer **all** questions.

Approved calculators are allowed.

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is **40**.

1 A community centre uses an online form to request visitor passes. The form contains the following fields:

- `VisitorName`, which must not be empty
- `PartySize`, which must be an integer
- `PassCode`, which must consist of the letter `V` followed by exactly three digits.

(a) For each field, state the most suitable validation check from the following list:

- format check
- presence check
- type check.

[3]

(b) Each of the following records is rejected. For each record, identify the validation rule that causes the rejection.

| Record | VisitorName | PartySize | PassCode |
|--------|-------------|-----------|----------|
| 1      | Nur         | four      | V318     |
| 2      | Wei Ming    | 4         | 318V     |

[2]

(c) A record containing `Alicia Tan`, `5` and `V381` passes all three validation checks. The visitor's actual party size is `3`.

Explain why the record is accepted even though it contains inaccurate data.

[2]

2 A radio system transmits short equipment-status labels as fixed-width binary values.

(a) Convert the 8-bit binary value `10101110` to hexadecimal. Show your working. [2]

(b) Convert the hexadecimal value `3D` to denary. Show your working. [2]

An extract from the character encoding used by the system is shown below.

| Character | Denary code | 8-bit binary code |
|-----------|-------------|-------------------|
| F         | 70          | 01000110          |
| R         | 82          | 01010010          |
| S         | 83          | 01010011          |
| T         | 84          | 01010100          |

- (c) Decode the following two bytes using the supplied extract:

01010010 01000110

[2]

- (d) Each character in a status label is represented by one byte. Calculate the storage, in bytes, required for a status label containing 18 characters. [1]
- (e) Explain why the sender and receiver must use the same character encoding. [1]

- 3 A public library operates several branches. Each branch has its own network. The branches share a private catalogue that can be accessed only by library staff.

- (a) State the most suitable network term for each description.
- The network within one library branch
  - The network that connects library branches in different locations
  - The private catalogue network that can be accessed only by authorised staff
- Choose from LAN, WAN and intranet. [3]
- (b) A staff tablet attempts to join a branch network for the first time.  
Describe how the switch and router allow the tablet to be registered on the network and assigned an IP address. [2]
- (c) The tablet is a client and the computer that stores the catalogue is a server.  
Explain the client-server interaction when a member of staff searches the catalogue. [2]
- (d) Explain why the tablet and catalogue server must use an agreed communication protocol. [2]

- 4 A maintenance department stores job codes in a list. The index of the first item is 0.

- (a) The following unsorted list is searched for the job code 392 using linear search:

[417, 203, 586, 144, 392, 251]

State, in order, the values compared with 392. State the index returned when the job code is found. [3]

- (b) The following two sublists have already been sorted into ascending order:

[144, 251, 417]      [203, 392, 586]

Show how the two sublists are merged into one sorted list. [3]

- (c) The following list is to be sorted into ascending order using merge sort:

[38, 12, 45, 7, 26, 19]

Draw diagrams to show all the splitting and merging stages. When a list contains an odd number of values, place the extra value in the left sublist. [3]

- (d) State the worst-case time complexity of merge sort using Big-O notation. [1]

5 A community centre stores course enrolments in the following flat file.

| EnrolmentID | ParticipantName | CourseCode | CourseTitle            | Venue    |
|-------------|-----------------|------------|------------------------|----------|
| E1041       | Aisha           | C12        | Digital Photography    | Studio 2 |
| E1042       | Ben             | C08        | Introduction to Coding | Lab 1    |
| E1043       | Aisha           | C08        | Introduction to Coding | Lab 1    |
| E1044       | Chen            | C15        | Robotics Basics        | Lab 3    |
| E1045       | Devi            | C08        | Introduction to Coding | Lab 1    |

- (a) State the name of the field that stores Studio 2.  
State all the values in the record with EnrolmentID E1044. [2]
- (b) Explain how storing course details in this flat file can lead to data redundancy and data inconsistency. [2]
- (c) Select a suitable single-field candidate key for the flat file.  
Explain why ParticipantName is not suitable as a candidate key. [2]

**END OF PAPER**